COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures & Algorithm	**SECTION:**	CPE21S1
DATE PERFORMED:	07/09/2026	**DATE SUBMITTED:**	07/14/2026
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ACTIVITY NO. HOA 1.1_ [ACTIVITY NAME] Basic C++ Programming

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

- PROCESS 1: **Create a function that will take two numbers and display the sum.**

SECTIONS	ANSWER
Header File Declaration Section	Proccess1.cpp > TwoSumNum(int, int) 1 //Alvarez, Joshua Caleb P. 2 #include <iostream>
Global Declaration Section	// Global Declaration int a, b;
Class Declaration and Method Definition Section	6 7 // Class Declaration & Method Declaration 8 int TwoSumNum(int a, int b); 9
Main Function	11 // Main Function 12 int main() { 13 14 std::cout << "Enter First number: "; 15 std::cin >> a; 16 std::cout << "Enter Second number: "; 17 std::cin >> b; 18 19 int total = TwoSumNum(a, b); 20 std::cout << "The sum is: " << total << std::endl; 21 return 0; 22 }

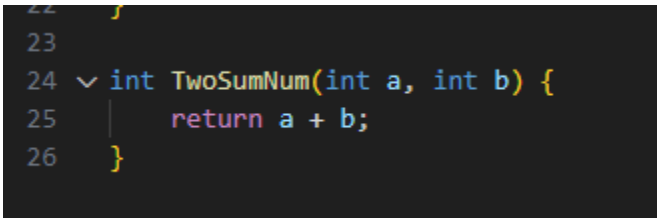
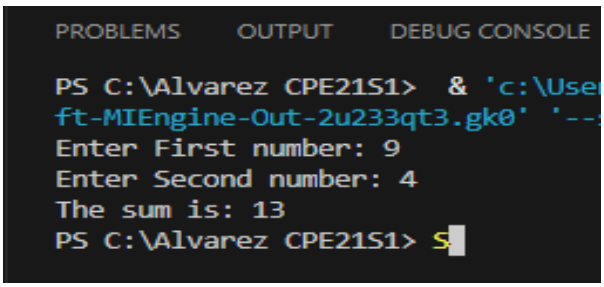
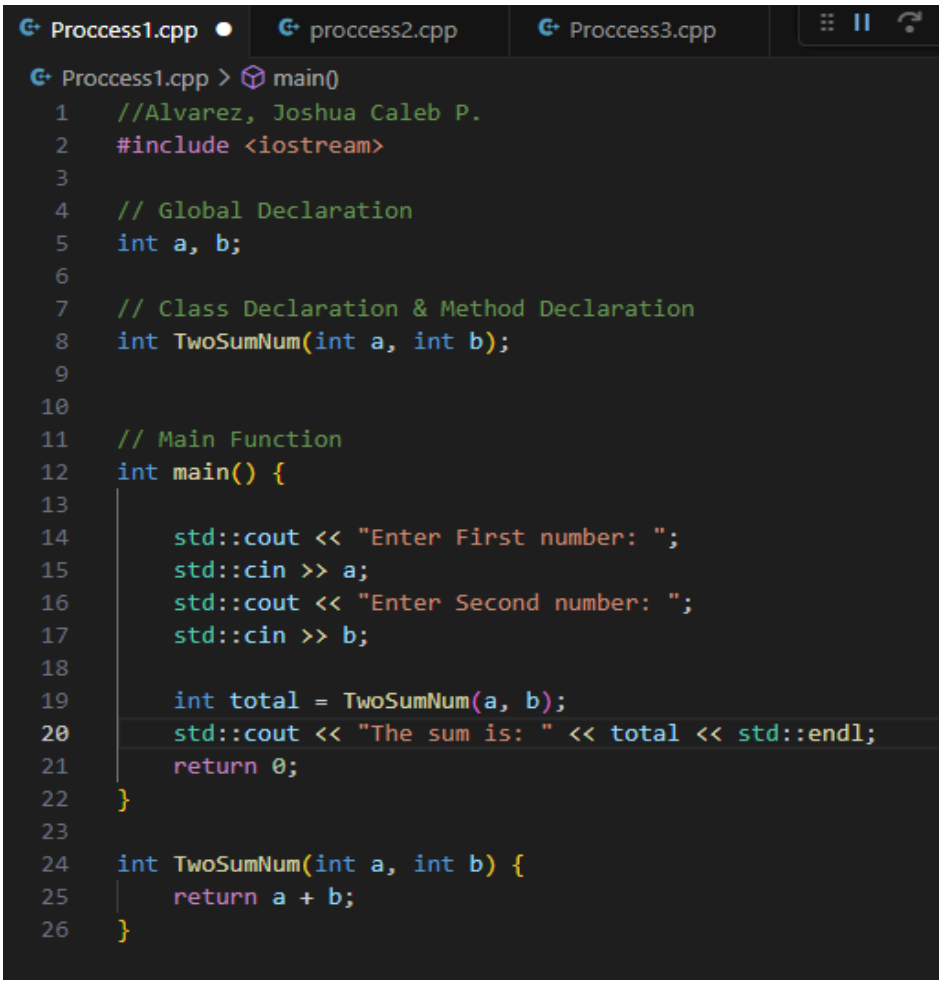
Method Definition	
OUTPUT	

TABLE 1-1 ILO A: C++ Structure Code for Answer

CODE FULL SCREENSHOT OF PROCESS 1



PROCESS 2: **Create a function that will return whether variable A is greater than variable B.**

SECTION	ANSWER
Header File Declaration Section	proccess2.cpp > ... 1 //Alvarez, Joshua Caleb P. 2 #include <iostream>
Global Declaration Section	3 4 int a, b; 5
Class Declaration and Method Definition Section	6 int NumResult(int a, int b); 7

Main Function	int main() {
	std::cout << "Enter First number: ";
	std::cin >> a;
	std::cout << "Enter Second number: ";
	std::cin >> b;
	return 0;
	}

Method Definition	18 int NumResult(int a, int b){
	19 if(a>b){
	20 std::cout << "The Number: "<< a <<" is greater than: " << b << std::endl;
	21 }else if(a<b){
	22 std::cout << " The Number: " << a << " is less than: " << b << std::endl;
	23 }
	24 return 0;
	25 }
	26

OUTPUT	Enter First Number: 45 Enter Second Number: 23 The Number 45 is greater than 23

TABLE 1-2 C++ Structure Code for Answer

CODE FULL SCREENSHOT FOR PROCESS 2

proccess2.cpp > ...

```cpp
//Alvarez, Joshua Caleb P.
#include <iostream>

int a, b;

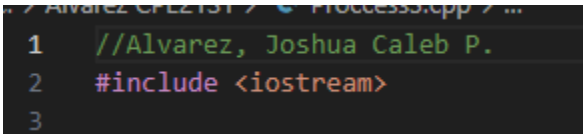
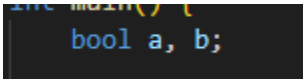
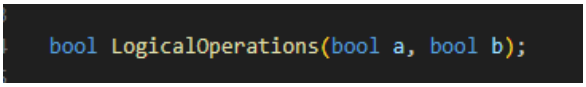
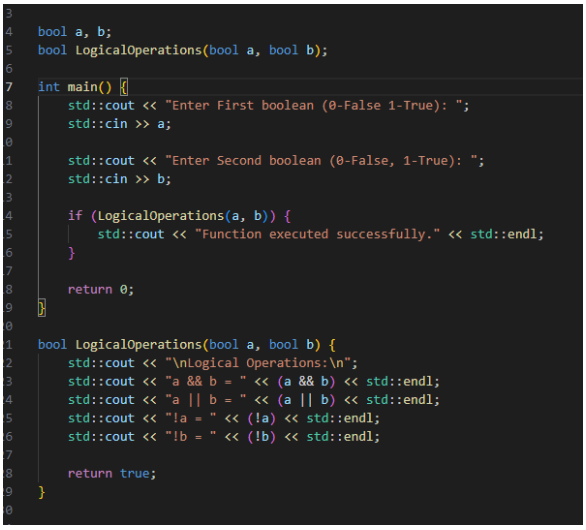
int NumResult(int a, int b);

int main() {
    std::cout << "Enter First number: ";
    std::cin >> a;
    std::cout << "Enter Second number: ";
    std::cin >> b;

    return 0;
}

int NumResult(int a, int b){
if(a>b){
    std::cout<<"The Number: "<<a<<"is greater than: "<< b<< std::endl;
}else if(a<b){
    std::cout<<"The Number: "<< a<<"is less than: "<<b<< std::endl;
}
    return 0;
}
```

PROCESS 3: **Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success**

SECTION	ANSWER
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	
Main Function	
Method Definition	

```
//Alvarez, Joshua Caleb P.
#include <iostream>
```

```
bool a, b;
```

```
bool LogicalOperations(bool a, bool b);
```

```
bool a, b;
bool LogicalOperations(bool a, bool b);

int main() {
    std::cout << "Enter First boolean (0-False 1-True): ";
    std::cin >> a;

    std::cout << "Enter Second boolean (0-False, 1-True): ";
    std::cin >> b;

    if (LogicalOperations(a, b)) {
        std::cout << "Function executed successfully." << std::endl;
    }

    return 0;
}

bool LogicalOperations(bool a, bool b) {
    std::cout << "\nLogical Operations:\n";
    std::cout << "a && b = " << (a && b) << std::endl;
    std::cout << "a || b = " << (a || b) << std::endl;
    std::cout << "!a = " << (!a) << std::endl;
    std::cout << "!b = " << (!b) << std::endl;

    return true;
}
```

```
}

bool LogicalOperations(bool a, bool b) {
    std::cout << "\nLogical Operations:\n";
    std::cout << "a && b = " << (a && b) << std::endl;
    std::cout << "a || b = " << (a || b) << std::endl;
    std::cout << "!a = " << (!a) << std::endl;
    std::cout << "!b = " << (!b) << std::endl;

    return true;
}
```

OUTPUT	Enter first Boolean value (0 or 1): 1 Enter second Boolean value (0 or 1): 0 AND: 0 OR : 1 NOT A: 0 NOT B: 1 Logical operations executed successfully.

TABLE 1-3 C++ Structure Code for Answer

Process 1 – Code Analysis

The program demonstrates the use of a user-defined function to add two integer values. It first asks the user to enter two numbers and stores them in global variables. The values are then passed to the `TwoSumNum()` function, which performs the addition and returns the sum to the `main()` function. Finally, the program displays the result on the screen. This activity shows how functions can simplify a program by separating the computation from the main program flow.

Process 2 – Code Analysis

This program uses a user-defined function to compare two integer values entered by the user. After receiving the input, the values are passed to the `NumResult()` function, which determines whether the first number is greater than or less than the second number using conditional statements (`if` and `else if`). The program then displays the appropriate message. This activity demonstrates how functions and decision-making statements can be combined to evaluate and compare user inputs.

Process 3 – Code Analysis

The program demonstrates the use of Boolean data types and logical operators in C++. It asks the user to input two Boolean values and passes them to the `LogicalOperations()` function. Inside the function, the program performs logical operations such as **AND (`&&`)**, **OR (`||`)**, and **NOT (`!`)**, displaying the result of each operation. The function returns `true` after successfully executing all logical operations, and the `main()` function confirms the successful execution by displaying a message. This activity helped demonstrate how Boolean expressions and functions work together to evaluate logical conditions in a program.

ILO A: Table 1-1 ILO A output observations and comments.

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

SECTION	ANSWER
Header File Declaration Section	`#include <iostream>`

Global Declaration Section	`public:`
	`  Triangle(double A, double B, double C);`
	`  void setAngles(double A, double B, double C);`
	`  const bool validateTriangle();`
	`};`

Class Declaration and Method Definition Section	`public:`
	`  Triangle(double A, double B, double C);`
	`  void setAngles(double A, double B, double C);`
	`  const bool validateTriangle();`
	`};`
	`Triangle::Triangle(double A, double B, double C) {`
	`  angleA = A;`
	`  angleB = B;`
	`  angleC = C;`
	`  totalAngle = A+B+C;`
	`}`
	`void Triangle::setAngles(double A, double B, double C) {`
	`  angleA = A;`
	`  angleB = B;`
	`  angleC = C;`
	`  totalAngle = A+B+C;`
	`}`

Main Function	`int main(){`
	`  //driver code`
	`  Triangle set1( A: 40, B: 30, C: 110);`
	`  if(set1.validateTriangle()){`
	`  std::cout << "The shape is a valid triangle.\n";`
	`  } else {`
	`  std::cout << "The shape is NOT a valid triangle.\n";`
	`  }`
	`  return 0;`
	`}`

Method Definition
 	
OUTPUT
	`const bool Triangle::validateTriangle() {`
	`  return (totalAngle <= 180);`
	`}`
	`The shape is a valid triangle.`

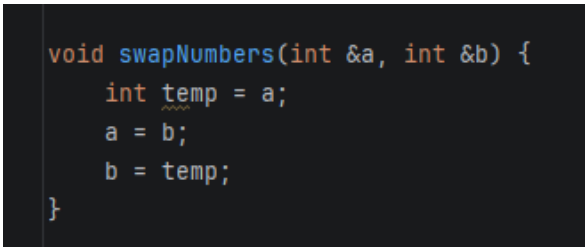
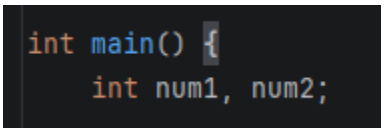
Table 1-1. ILO B C++ Structure Code for Answer

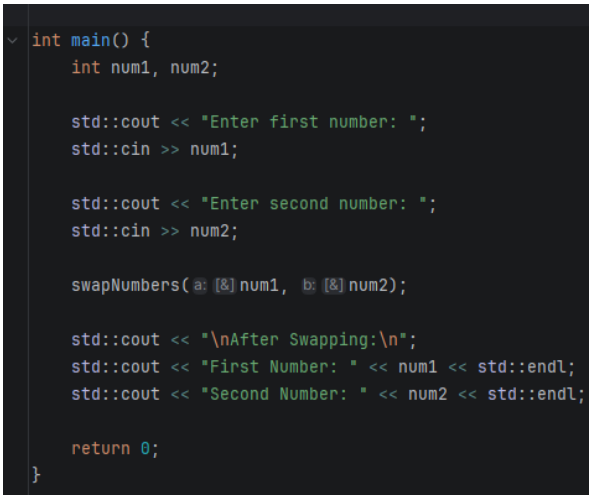
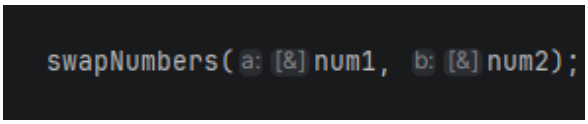
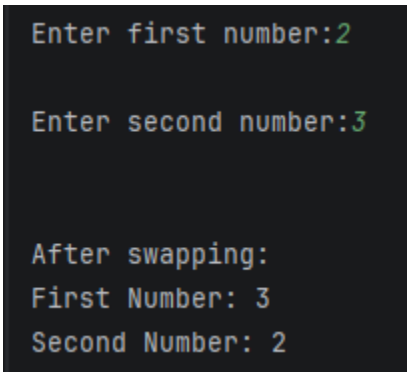
My observations is the program successfully validated whether the given angles formed a triangle. Using the input values of **40°, 30°, and 110°**, the program calculated the total angle as **180°** and displayed the message **"The shape is a valid triangle."** This confirms that the constructor correctly initialized the object and that the `validateTriangle()` member function worked as intended. The program also demonstrated the use of object-oriented programming concepts such as classes, constructors, member functions, and Boolean return values. Overall, the output matched the expected result, showing that the program was able to determine the validity of the triangle based on the sum of its interior angles.

Table 1-2. ILO B output observations and comments.

2.ANSWERS FROM THE SUPPLEMENTARY ACTIVITY

ILO C: Solve Different Problems using the C++ Programming Language
1.Create a C++ program to swap the two numbers in different variables.

SECTION	ANSWER
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	

Main Function	
Method Definition	
OUTPUT	

```
int main() {
    int num1, num2;

    std::cout << "Enter first number: ";
    std::cin >> num1;

    std::cout << "Enter second number: ";
    std::cin >> num2;

    swapNumbers(a: [&]num1, b: [&]num2);

    std::cout << "\nAfter Swapping:\n";
    std::cout << "First Number: " << num1 << std::endl;
    std::cout << "Second Number: " << num2 << std::endl;

    return 0;
}
```

```
swapNumbers(a: [&]num1, b: [&]num2);
```

```
Enter first number:2

Enter second number:3


After swapping:
First Number: 3
Second Number: 2
```

Table 1-1. ILO C: C++ Structure Code for Answer

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

SECTION	ANSWER
Header File Declaration Section	`#include <iostream>` `#include <cmath>`

Global Declaration Section	`double kelvinToFahrenheit(double kelvin) {`
	`  return (kelvin - 273.15) * 9 / 5 + 32;`
	`}`

Class Declaration and Method Definition Section	`double kelvin;`

Main Function	`int main() {`
	`  double kelvin;`
	`  std::cout << "Enter temperature in Kelvin: ";`
	`  std::cin >> kelvin;`
	`  std::cout << "Temperature in Fahrenheit: "`
	`  << kelvinToFahrenheit(kelvin) << std::endl;`
	`  return 0;`
	`}`

Method Definition
 	
OUTPUT
	`<< kelvinToFahrenheit(kelvin) << std::endl;`
	
	`Enter temperature in Kelvin:23`
	
	`Temperature in Fahrenheit: -418.27`

Table 1-2. ILO C: C++ Structure Code for Answer

3.Create a C++ program that has a function that will calculate the distance between two points.

SECTION	ANSWER
Header File Declaration Section	`#include <iostream>` `#include <cmath>`
Global Declaration Section	`double calculateDistance(double x1, double y1, double x2, double y2)`
Class Declaration and Method Definition Section	`return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));` `}`

Main Function	`int main() {`
	`double x1, y1, x2, y2;`
	
	`std::cout << "Enter x1: ";`
	`std::cin >> x1;`
	
	`std::cout << "Enter y1: ";`
	`std::cin >> y1;`
	
	`std::cout << "Enter x2: ";`
	`std::cin >> x2;`
	
	`std::cout << "Enter y2: ";`
	`std::cin >> y2;`
	
	`std::cout << "\nDistance = "`
	`<< calculateDistance(x1, y1, x2, y2) << std::endl;`
	
	`return 0;`
	`}`

Method Definition
 	
OUTPUT
	`std::cout << "\nDistance = "`
	`<< calculateDistance(x1, y1, x2, y2) << std::endl;`
	
	`Enter x1:5`
	
	`Enter y1:4`
	
	`Enter x2:6`
	
	`Enter y2:5`

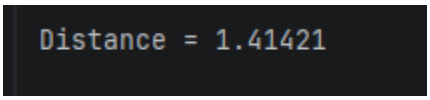
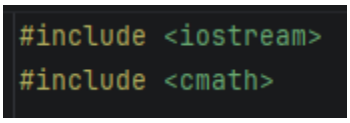
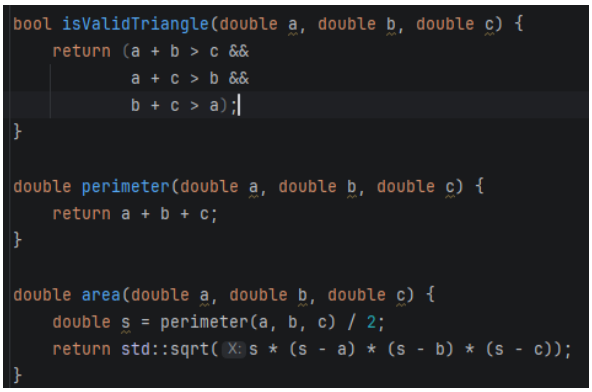
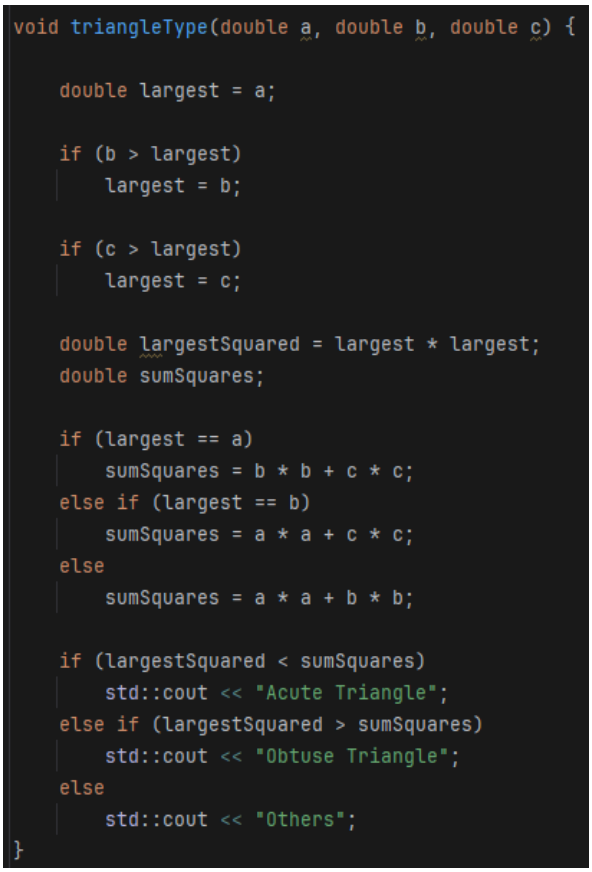
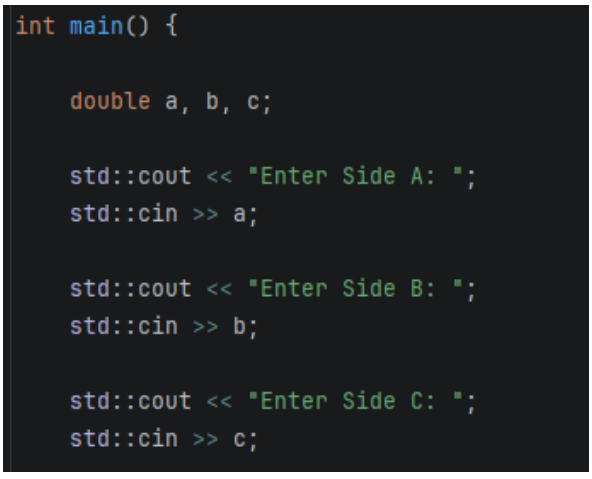
	

Table 1-3. ILO C: C++ Structure Code for Answer

4.Modify the code given in ILO B and add the following functions

a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

SECTION	ANSWER
Header File Declaration Section	
Global Declaration Section	

Class Declaration and Method Definition Section	
Main Function	

```
void triangleType(double a, double b, double c) {

    double largest = a;

    if (b > largest)
        largest = b;

    if (c > largest)
        largest = c;

    double largestSquared = largest * largest;
    double sumSquares;

    if (largest == a)
        sumSquares = b * b + c * c;
    else if (largest == b)
        sumSquares = a * a + c * c;
    else
        sumSquares = a * a + b * b;

    if (largestSquared < sumSquares)
        std::cout << "Acute Triangle";
    else if (largestSquared > sumSquares)
        std::cout << "Obtuse Triangle";
    else
        std::cout << "Others";
}
```

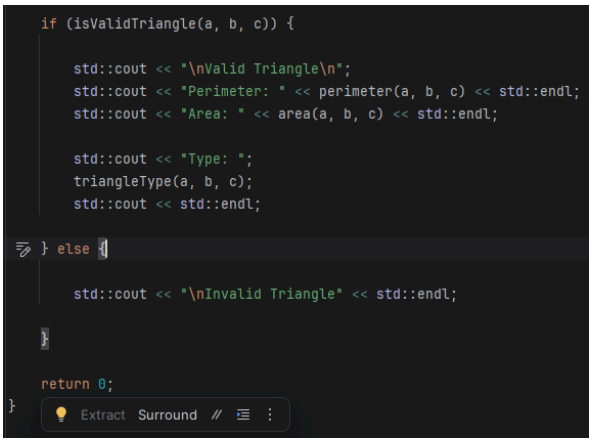
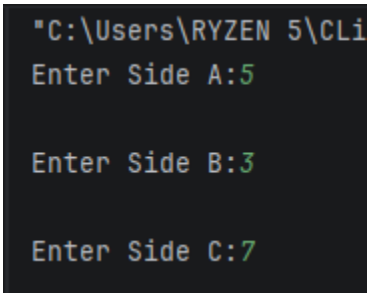
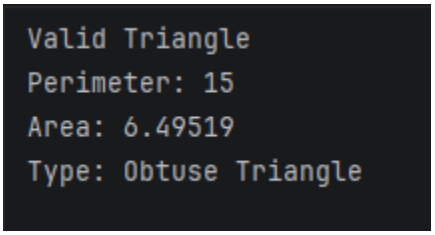
```
int main() {

    double a, b, c;

    std::cout << "Enter Side A: ";
    std::cin >> a;

    std::cout << "Enter Side B: ";
    std::cin >> b;

    std::cout << "Enter Side C: ";
    std::cin >> c;
```

Method Definition	
OUTPUT	 

```
    if (isValidTriangle(a, b, c)) {

        std::cout << "\nValid Triangle\n";
        std::cout << "Perimeter: " << perimeter(a, b, c) << std::endl;
        std::cout << "Area: " << area(a, b, c) << std::endl;

        std::cout << "Type: ";
        triangleType(a, b, c);
        std::cout << std::endl;

    } else {

        std::cout << "\nInvalid Triangle" << std::endl;

    }

    return 0;
}
```

Extract Surround //

```
"C:\Users\RYZEN 5\CLi
Enter Side A:5

Enter Side B:3

Enter Side C:7
```

```
Valid Triangle
Perimeter: 15
Area: 6.49519
Type: Obtuse Triangle
```

Table 1-4 IILO C: C++ Structure Code for Answer

Task 1 – Swap Two Numbers Using a Function (Code Analysis)

This program uses a function to swap the values of two numbers entered by the user. The numbers are passed to the `swapNumbers()` function, where a temporary variable is used to exchange their values. After the function finishes, the swapped numbers are displayed. This program shows how functions can perform a specific task and make the code easier to understand.

Task 2 – Kelvin to Fahrenheit Conversion (Code Analysis)

This program uses a function to convert a temperature from Kelvin to Fahrenheit. The user enters a temperature in Kelvin, and the value is passed to the `kelvinToFahrenheit()` function. The function performs the conversion using the correct formula and returns the result. The converted temperature is then displayed. This activity shows how functions can be used to solve mathematical problems in programming.

Task 3 – Distance Between Two Points (Code Analysis)

This program calculates the distance between two points using a function. The user enters the coordinates of two points, and the values are passed to the `calculateDistance()` function. The function uses the distance formula to compute the answer and returns the result. The distance is then displayed on the screen. This activity helped me understand how functions can be used to perform calculations and organize a program.

Task 4 – Triangle Validation, Area, Perimeter, and Classification (Code Analysis)

This program uses different functions to perform several operations on a triangle. It first checks if the given side lengths can form a valid triangle. If the triangle is valid, the program calculates its perimeter and area using separate functions. It also identifies whether the triangle is acute, obtuse, or others based on its side lengths. Using separate functions makes the program easier to read, understand, and manage.

Table 1.5 ILO C: Code Analysis & output observations

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Conclusion

Summary of Lessons Learned

In this laboratory activity, I learned how to create and use functions in C++ to solve different programming problems. I also learned how to pass values to functions, return values, and organize a program into smaller and more manageable parts. The activities helped me understand the importance of functions in making programs easier to read, debug, and reuse.

Analysis of the Procedure

The procedure was easy to follow because each activity focused on one specific concept. I started by analyzing the given problem, planning the solution, writing the code, and testing the program to check if the output was correct. I encountered a few errors while coding, such as syntax mistakes and logic errors, but I was able to correct them by reviewing my code and testing it several times. This process improved my problem-solving and debugging skills.

Analysis of the Supplementary Activity

The supplementary activities provided additional practice in applying functions to different situations, such as swapping values, converting temperatures, calculating distances, and solving triangle-related problems. These exercises challenged me to think logically and apply mathematical formulas in programming. Completing these activities also improved my confidence in writing C++ programs independently.

Concluding Statement / Feedback

Overall, I believe I did my best in completing this laboratory activity. I was able to understand how functions work and apply them to different programming problems. However, I encountered a challenge while using Visual Studio Code. My programs could not run because my project contained multiple C++ files with more than one `main()` function, which caused compilation errors. Because of this issue, I was unable to properly save and run my `.cpp` files, and I was not able to submit the required C++ source files on time. Although this affected my submission, I learned the importance of properly organizing project files and understanding how development environments work. Moving forward, I will continue practicing C++ programming and improve my debugging skills so I can solve similar problems more efficiently.

COMPUTER ENGINEERING

QUEZON CITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **CHAT GPT, CLAUDE AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **IDEA GENERATE , CODE ASSISTANCE , ESSAY SUMMARIZATION**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was used to help me understand some C++ concepts, fix coding errors, and improve my documentation. I still wrote, reviewed, tested, and completed the required work on my own.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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